

Peekaboo: A Smart Object for Enhancing Parent–Child Eye Glances During Car Rides

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Fig. 1. Peekaboo: A TUI to encourage eye glance between parent and child during a car ride

Parent–child communication is vital for children’s emotional, social, and cognitive development, often emerging through shared activities and spontaneous interaction. Car rides offer a unique opportunity for such moments, yet are increasingly disrupted by mobile phones that shift attention inward. We explore the possibility of promoting parent–child communication by facilitating brief moments of mutual eye glancing, a behavior shown to activate neural systems that enhance social interaction. Peekaboo is a smart object mounted in front of the passenger seat, combining a mirror, an electric motor, and a camera. Through subtle, non-intrusive movements, the mirror gently invites the passenger to lift their gaze and establish eye contact with the driver. Designed through a human-centered process, the interaction is intentionally brief and non-demanding. Our study demonstrate the potential of minimal gestures to encourage brief eye glancing and support parent-child communication during car rides.

CCS Concepts: • **Human-centered computing** → **Interaction design**; *Empirical studies in HCI*.

Additional Key Words and Phrases: Parent-child communication, smart object, glance, eye contact, car rides

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Manuscript submitted to ACM

Manuscript submitted to ACM

ACM Reference Format:

Noa Morag Yaar, Ofir Sadka, Ido Vigodman, Amit Berenstein Goldenberg, Almog Arazi, Oren Zuckerman, and Hadas Erel. 2026. Peekaboo: A Smart Object for Enhancing Parent–Child Eye Glances During Car Rides. In *Proceedings of Interaction Design and Children (IDC '26)*. ACM, New York, NY, USA, 8 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Parent-child communication is important for the development of a child's emotional, social, and cognitive development [21, 38, 43]. Parents play a central role in the establishment of healthy communication with their children, which has a profound effect ranging from a child's well-being and self-esteem from an early age [45], to social functioning and academic performance [15, 16]. Healthy communication comes from shared activities and meaningful conversations [13, 19, 35, 41]. During social interactions, eye contact plays a central role in signaling willingness to engage, demonstrating attentiveness, and cueing the continuation of conversation [11, 32–34]. It has also been shown to directly activate brain systems associated with strong emotional responses, thereby influencing perceptual and cognitive processing [27, 29]. More broadly, eye contact shapes interpersonal dynamics and emotional connection, serving as a powerful facilitator of communication [3, 30].

However, while on their phone, parents and children miss the opportunity for healthy communication. Studies show how screens, especially mobile phones, have become a significant challenge for healthy communication between parents and children, in particular by distracting their attention, causing disengagement from present interactions [4, 12, 22], leading to emotional unavailability, reduced engagement [26, 37, 47] and lack of eye contact which prevents interaction. This becomes even more difficult during adolescence [7, 23, 26].

The context of family car rides offers a unique opportunity for meaningful communication between parents and children. Driving children during the week is a familiar routine for many parents, often along with work and other commitments [25, 40]. The car can also serve as a dynamic social space, offering opportunities for interactions beyond transportation [10, 28, 36]. Its intimate setting creates opportunities for meaningful conversations that parents and children often miss, while the car itself provides a neutral, supportive space [49]. Furthermore, the structured nature of a car journey, with a clear beginning, middle, and end, makes it a unique context for sharing feelings, personal stories, and perspectives, fostering meaningful communication between parents and children [9, 36].

This paper explores the potential of enhancing parent–child communication during car rides by introducing opportunities for eye contact between them. We investigate how a smart object can gently encourage glances and, ultimately, moments of eye contact to foster more meaningful interactions between parents and children in the car. We present a design study and early stage prototype of Peekaboo, a smart object positioned on the dashboard in front of the passenger seat, consisting of a mirror, an electric motor, and a camera. While the car is in motion, Peekaboo tracks whether the passenger is looking forward. If the system does not detect the passenger's eyes, it performs a subtle, brief movement of the mirror to attract the attention of the passenger without distracting the driver. When the child lifts their gaze, the mirror aligns to reflect the driver's face, enabling a moment of eye contact between parent and child.

2 Related Work

Related works include technological interventions to promote parent-child interaction in car rides, and embodied technological interventions aimed at encouraging social interactions.

2.1 Technological interventions to promote parent-child interaction in car rides

Various technological interventions have focused on enhancing parent-child interaction during car rides. For example, Mileys is a location-based, augmented reality collecting game, where children discover virtual characters tied to real-world places during car rides. The game presents an interactive experience that strengthens family connection, encourages awareness of surroundings, and makes journeys more engaging and educational [49]. Another example is "RiddleRide," a design concept for a multiplayer quiz app that offers an entertaining and educational experience for backseat passengers. The system delivers voice riddles through the built-in sound system, with each question read aloud, enabling all passengers to take part and allowing children who cannot read yet to join in the fun [46]. Another example is BlokCar is a location-based in-car game that encourages passengers to engage with the outside environment through audio cues that guide them in completing location-based missions using handheld devices. The system is configured by parents via a mobile application prior to the journey, which tailors interactive content based on the child's preferences, route, and driving conditions, and includes features such as an augmented reality interface and tangible blocks that enable screen-free interaction with the car's media system [48]. These studies highlight the potential of car rides to foster meaningful family interactions, emphasizing the importance of thoughtful design to ensure positive user experiences while minimizing distractions. In this work, we leverage technologies that promote eye contact, enabling an unstructured and open-ended communication between parents and children while in the car.

2.2 Embodied technological interventions aimed at encouraging social interactions

Studies have leveraged humans' natural sensitivity to social cues to support social interaction by shaping nonverbal behaviors. For example, Sadka et al. (2021) studied whether autonomous everyday furniture could reduce awkwardness and support positive opening encounters between strangers. They found that robotic bar stools rotating people toward each other increased social interaction and interpersonal communication [42]. Morag et al. (2020) introduced a table spinning top designed to minimize mobile phone distractions during family dinner [39]. When the spinning top detects phone usage, it begins to spin, and family members must touch the spinning top together to make it stop. The study demonstrated how this activity encouraged communication between family members. In the context of public spaces, Balesrini et al. (2016) presented the Jokebox, a technological prototype that requires two passers-by strangers to coordinate actions to hear a joke. The technology encouraged strangers to establish eye contact and often resulted in follow up interactions [5]. Peekaboo builds on these approaches by gently invite a glance to establish an eye contact between parents and children. Instead of introducing additional content or activities, it leverages minimal intervention to support spontaneous, open-ended communication, aligning with the unique constraints and opportunities of the car environment.

3 Design

The design process described below follows Design Thinking and Human Centered Design (HCD) approach [8, 14, 24]. We began by identifying our users' needs and validating them through interviews and consultations with relevant experts. We then proceeded to an iterative design process that involved rapid prototyping and user testing.

3.1 Need Study

Our need study included three experts; a developmental psychologist, a psychologist, and a speech therapist specialized in parent-child communication. The study also includes four parents and five children (aged 9-12). The interviews

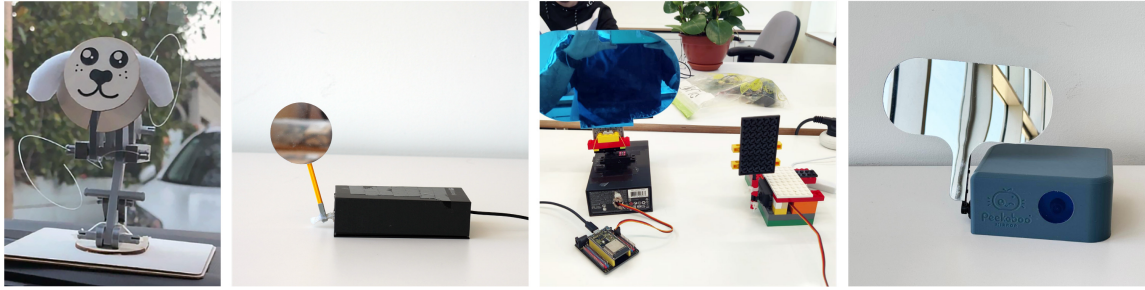


Fig. 2. Sample of rapid prototyping from left to right: Testing the concept of a dog-like object in a car setting, testing movement and sizes of mirrors, and the overall design of Peekaboo.

included open questions to understand the opportunities and challenges of parents-child interaction. The three experts revealed that in therapy parents often complain about how their children are disconnected from their social environment. The developmental psychologist shared how screens are present all the time: "It's hard when a child is used to eat in front of their phone, or fall asleep with their phone". The speech therapist emphasized how healthy communication between parents and children is compromised: "If we want the child to learn how to communicate with others, we need to remember to make eye contact and really be with them". Following the experts we interviewed parents and children to learn more about their interaction in the home. A parent said: "Once they are on their phone, they don't hear or see anything else" [P2], "I really want him to let go of the phone but it is very hard" [P1], "They become passive and it is very hard to engage with them" [P3] or "even when together it is hard for us to spend time with them" [P1]. Children said: "I ask my mom to get off the phone as well but she says it's for work" [P4], "Mom, you are always on the phone!" [P5]. The decision to focus on a driving context came as inspiration from one of the parents: "I remember being shy and only sharing when in the car on the way to an afternoon activity, it was just me and my mom in the car" [P4]. We further examined the context of family car rides and found that our interview insights align with findings from previous research. Studies highlight the potential of car rides as opportunities for quality time [10, 28, 36]. Parents often drive their children to and from weekly activities, typically in short trips with one or two children, creating a setting that supports social interactions [6, 36, 40].

Our next step in the design process was to map the car ride journey to identify potential moments for intervention. Although greetings have been shown to promote healthy communication [1, 17, 18, 30, 31], we found that the initial entry into the car is often too hectic to effectively introduce an intervention. We therefore chose to focus on the ride itself. Prior research highlights the importance of eye contact in social interaction and suggests that even brief glances, defined as momentary looks or gaze shifts, can help initiate it [2, 44]. Based on this, we derived the following design guidelines: (1) promote "mobile-free" time to enhance the quality of parent-child communication in the car, (2) use minimal, non-intrusive cues that support communication without imposing on it, and (3) facilitate eye glances as catalysts for conversation.

3.2 Iterative Prototype Design

Early prototypes explored various form factors and feedback mechanisms. An initial playful, figurative object was chosen for its familiarity in cars (see Fig. 2, left), but was found to be overly distracting and to introduce an unintended playful tone that conflicted with the intimate communication we aimed to support. Subsequent iterations focused on



Fig. 3. User testing Peekaboo from left to right: Checking the angle of mirror in front of the passenger seat, participants test the movement of the mirror in a pretend driving context, demoing the final Peekaboo in a lab setting.

minimizing movement and reducing visual salience, drawing on prior work demonstrating how subtle nonverbal cues can meaningfully shape social interaction [39, 42]. An attempt to provide feedback through continuous movement during conversation, informed by behavior change literature [20], proved too intrusive and disrupted natural communication.

We then explored the use of a mirror as a means to facilitate eye glance between passenger and driver. Inspired by rear-facing infant car mirrors (thus the name *Peekaboo*), we investigated a front-mounted mirror designed to encourage brief glances without requiring direct turning of the head (see Fig. 2). Next, in a lab setting, with We simulated an in car interaction with 10 university students (ages 20–27) (see Fig. 3, middle). We tested the mirror size, angle, and movement direction. Participants preferred a short vertical motion, which was perceived as noticeable yet least distracting. Further iterations refined the placement of the device on the dashboard, positioning it in front of the passenger to minimize driver distraction while maintaining engagement. We then tested for timing and duration of the intervention which helped us finalize that activating the mirror will happen once, approximately one minute into the drive for best notice-ability and the least intrusiveness. These iterations informed the design of the mid-fidelity prototype described in the following section.

4 Mid fidelity prototype

The mid-fidelity prototype consists of a mirror, an electric motor, and a camera mounted on the dashboard in front of the passenger seat and directed toward the passenger. A computer vision algorithm processes the camera input to detect the presence of the child’s pupils. After approximately one minute into the ride, if the system does not detect the child’s eyes, the mirror performs two brief vertical movements to subtly draw the child’s attention. When the child lifts their gaze, the mirror remains steady at an angle that enables an eye glance between the parent and the child. Peekaboo is designed to operate within the driver’s natural line of sight and to perform the gesture only once per journey. By relying on minimal movement and limited intervention, the system aims to encourage moments of connection without imposing on interaction or introducing unnecessary distraction.

4.1 System Logic

To support the interaction described above, Peekaboo relies on a lightweight vision-based system that detects the presence of the passenger’s gaze rather than continuously tracking attention. The system remains passive for most of

the car ride, activating only when prolonged visual disengagement is detected. Once triggered, it performs a single, brief mirror movement before returning to an idle state, ensuring the interaction stays minimal and non-intrusive. This design reflects our goal of limiting system intervention while enabling meaningful moments of connection through a simple, embodied cue.

5 Discussion

Peekaboo was designed to enhance eye contact between parents and children during car rides, drawing on HCI and psychology research that emphasizes the importance of parent-child communication and the role of eye contact in supporting social interaction. Our design process identified mobile phones as a primary source of distraction, limiting opportunities for meaningful engagement, while also revealing car rides as an underutilized setting for shared interaction. Designing for this context required careful consideration of its unique constraints, particularly safety and the dynamic nature of attention during driving, as well as identifying appropriate moments for intervention. The resulting design employs minimal, non-intrusive cues that invite rather than impose interaction, using brief mirror movements to facilitate natural eye glances as catalysts for conversation. This approach balances engagement with safe driving practices, fostering moments of connection without disrupting the driving task. Future work will further explore the system's integration within a dedicated safe-driving mode and evaluate its impact in real-world settings.

6 Limitation

Car rides require careful consideration to ensure that technology does not compromise driving safety. Although diverting the gaze is already well integrated into driving habits, we recognize the fact that creating eye contact with a passenger differs fundamentally from checking side and rear mirrors. The design of Peekaboo minimizes risk by operating within the driver's natural line of sight and promoting only brief, momentary glances. These short interactions have been shown to be meaningful, yet they require further controlled testing to fully assess their impact on driver behavior and safety. Future evaluation should include parents and children to examine how Peekaboo's movement influences in-car interaction, building on the current adult-based prototyping that focused on safety, timing, and non-intrusiveness.

7 Selection of Children and Participation

This design study involved five children aged 9–12. All participants were included only after their parents signed a consent form and confirmed that their children were willing to take part. The consent form explained that all collected data would remain anonymous and used solely for research purposes. Before each interview, researchers checked with both the parents and the children to ensure they felt comfortable participating, emphasizing that there were no right or wrong answers and that all feedback was valuable. The study received ethical approval from the University, and all data were stored securely and shared anonymously only with authorized research personnel.

8 Conclusion

This work explores how minimal, non-intrusive gestures of a smart object can encourage eye glance between parents and children during car rides. Situated within the constraints of the driving context, this design study highlights the potential of subtle interventions to support moments of connection in everyday family interactions.

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